Solving $AX=b$, RREF

$$x_1 + 2x_2 + 2x_3 + 2x_4 = b_1$$
$$2x_1 + 4x_2 + 6x_3 + 8x_4 = b_2$$
$$3x_1 + 6x_2 + 8x_3 + 10x_4 = b_3$$

$$\left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 2 & 4 & 6 & 8 & b_2 \\ 3 & 6 & 2 & 10 & b_3 \end{array}\right]$$

Augmented matrix $= [A\ b]$

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 4 & b_2 - 2b_1 \\ 0 & 0 & 2 & 4 & b_3 - 3b_1 \end{array}\right]$$

pivot columns

$$\rightarrow \left[\begin{array}{cccc|c} 1 & 2 & 2 & 2 & b_1 \\ 0 & 0 & 2 & 6 & b_2 - 2b_1 \\ 0 & 0 & 0 & 0 & b_3 - b_2 - b_1 \end{array}\right]$$

$$0 = b_3 - b_2 - b_1$$

$$\underline{\underline{\text{OK } b = \begin{bmatrix} 1 \\ 5 \\ 6 \end{bmatrix}}}$$

Solvability condition on b:

$Ax=b$ solvable whe b is in $C(A)$

If a comb of rows of A gives zero row then the sam comb of entries of b must give 0

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to find complete Sol to $Ax=b$

① $X_{particilar}$: set all free variables to 0 then solve $Ax=b$ for pivot variables

$$x_1 + 2x_3 = 1 \quad , \quad x_1 = -2$$
$$2x_3 = 3 \quad , \quad x_3 = 3/2$$

$$x_p = \begin{bmatrix} -2 \\ 0 \\ 3/2 \\ 0 \end{bmatrix}$$

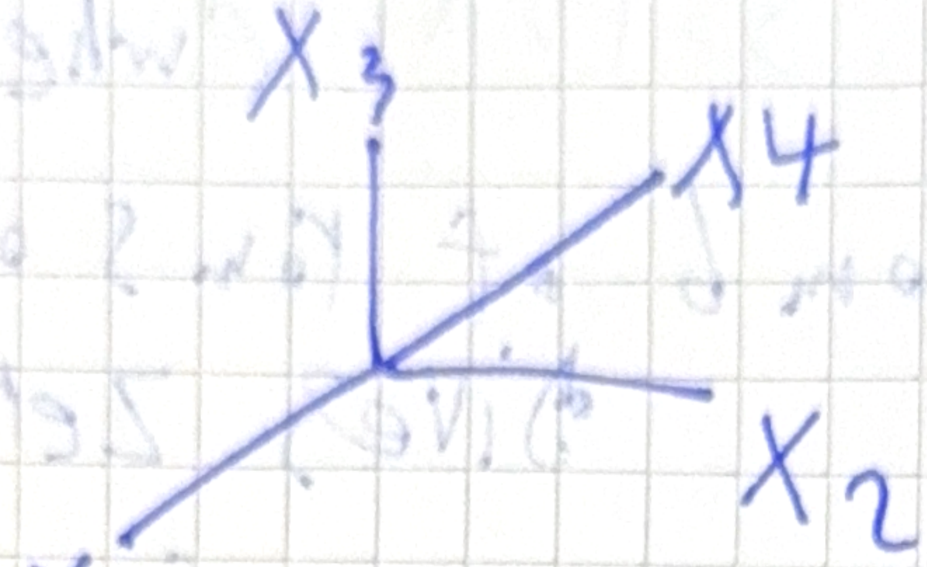
② null space

$X = X_p + X_n$

$$\begin{cases} A x_p = b \\ A x_n = 0 \end{cases}$$

$$A(x_p + x_n) = b$$

Plot null solutions $x \in \mathbb{R}^4$



$$x = x_p + x_n$$

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m by n matrix A of rank r

(Know $r \leq m$, $r \leq n$)

Full column rank means $r = n$

$\therefore$ No free variables

$N(A) = \{\text{zero vectors}\}$ solutions to $Ax = b$. $x = x_p$ unique solution if it exists (0 or 1 solution)

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 1 \\ 6 & 1 \\ 5 & 1 \end{bmatrix}, \quad R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$$

Full row rank means $r = m$

can solve $Ax = b$ for every b

Left with $n - r$ free variables (Exist)

$n - m$

$$A = \begin{bmatrix} 1 & 2 & 6 & 5 \\ 3 & 1 & 1 & 1 \end{bmatrix}$$

$$R = \begin{bmatrix} 1 & 0 & & \\ 0 & 1 & F & \\ & & & \end{bmatrix}$$

$r = m = n$

$R = I$

1 solution

$r = m < n$

$R = \begin{bmatrix} I \\ 0 \end{bmatrix}$

(0 or 1 solution)

$r = m < n$

((~~∞~~ solutions))

$r < m, r < n$

$R = \begin{bmatrix} I & F \\ 0 & 0 \end{bmatrix}$

((0 or ∞ solutions))